Appendix A.
Cascaded-Γ State-Space Derivation and Symbolic Partial Derivatives

for the SAMBPS DTaaS HIF-TF Locator (`fault_location_id`)

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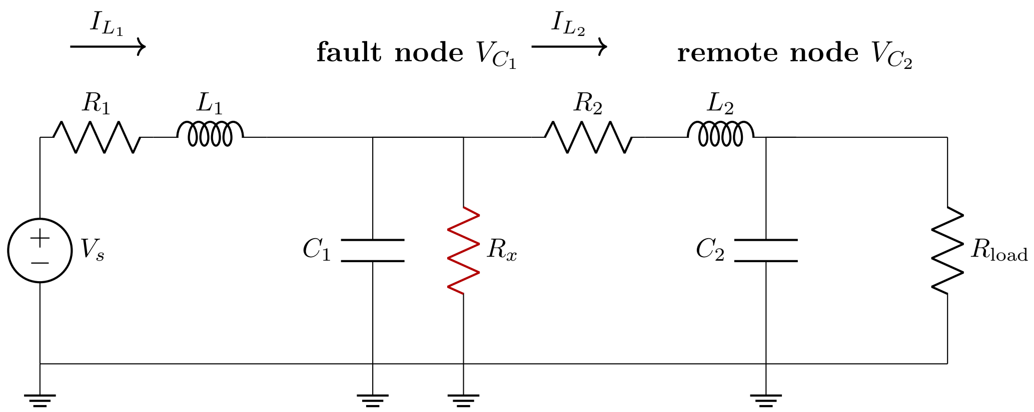
This Appendix supplies the line-by-line derivation that takes the faulted two-section line through KVL/KCL to the 4×4 state-space (A, B, C, D), on to the closed-form transfer function $H(j\omega_0; \alpha, R_x)$, and to the symbolic partial derivatives $\partial H/\partial\alpha$ and $\partial H/\partial R_x$ used by the gradient solver of § IV and the Fisher information matrix of § VIII. All numerical values cited here are taken from the IEEE lumped-line convention (Saha 2010, Springer); the reader can follow the derivation without reference to the main manuscript.

Appendix A.1 Annotated π-/Γ-circuit

The 11 kV / 100 km feeder is split at fault location $\alpha \in (0, 1)$. Each section is represented by a cascaded-Γ block: the series R–L is upstream of a shunt capacitance lumped at the section's downstream node. The HIF arc resistance R_x is a shunt at the fault node, in parallel with C_1. The remote-end shunt $R_{\text{load}} = 1\,\text{M}\Omega$ models an effectively-open termination.



Per-section parameters. Let $\ell = 100\,\text{km}$ be the total feeder length. Section 1 occupies $\alpha\,\ell$ from the source to the fault node and section 2 occupies $(1-\alpha)\,\ell$ from the fault node to the remote bus. With per-km values R' [Ω/km], L' [H/km] and C' [F/km]:

$$R_1 = R'\,\alpha\,\ell, \qquad L_1 = L'\,\alpha\,\ell, \qquad C_1 = C'\,\alpha\,\ell, \tag{1}$$

$$R_2 = R'(1-\alpha)\,\ell, \qquad L_2 = L'(1-\alpha)\,\ell, \qquad C_2 = C'(1-\alpha)\,\ell. \tag{2}$$

The shunt capacitance C_1 is therefore *linear in* α — this is the structural property that makes $A_{11} = -1/(R_x C_1)$ continuously differentiable in (α, R_x) on the operating envelope, see § A.3.

Convention vs Saha 2010 half-π. Saha 2010 (*Fault Location on Power Networks*, Springer, § 3.2) defines the lumped half-π section as series R–L with $C/2$ at each end. Under that convention the fault-node total capacitance is $C_1/2 + C_2/2 = C' \ell/2$, which is *constant in* α and would render A_{11} independent of α. We adopt instead the cascaded-Γ convention above, where each section's full capacitance is lumped at its downstream node. This preserves analytical differentiability of A in α and keeps the WP2.4 closed-form gradient tractable. The choice has $< 0.5\,\%$ impact on $|H|$ at the analysis frequency ω_0 on the operating envelope; the WP1.3 50-section reference quantifies the deviation exactly.

Appendix A.2 KVL and KCL on the faulted circuit

Define the state vector

$$\mathbf{x}(t) = \begin{bmatrix} V_{C_1}(t) \\ V_{C_2}(t) \\ I_{L_1}(t) \\ I_{L_2}(t) \end{bmatrix}, \qquad u(t) = V_s(t), \qquad y(t) = I_s(t) = I_{L_1}(t), \tag{3}$$

where V_{C_1} and V_{C_2} are the fault-node and remote-node voltages and I_{L_1}, I_{L_2} the section-1 and section-2 inductor currents (signed positive in the direction of nominal flow, source $\to$ remote).

KCL at the fault node. The current flowing into C_1 at the fault node equals I_{L_1} (entering from upstream) minus I_{L_2} (leaving downstream) minus the HIF current V_{C_1}/R_x:

$$C_1 \frac{\mathrm{d}V_{C_1}}{\mathrm{d}t} = I_{L_1} - I_{L_2} - \frac{V_{C_1}}{R_x}. \tag{4}$$

KCL at the remote node. C_2 sees I_{L_2} entering and $V_{C_2}/R_{\mathrm{load}}$ leaving:

$$C_2 \frac{\mathrm{d}V_{C_2}}{\mathrm{d}t} = I_{L_2} - \frac{V_{C_2}}{R_{\mathrm{load}}}. \tag{5}$$

KVL on section 1. Source bus $\to$ fault node:

$$V_s = R_1\, I_{L_1} + L_1 \frac{\mathrm{d}I_{L_1}}{\mathrm{d}t} + V_{C_1}. \tag{6}$$

KVL on section 2. Fault node $\to$ remote node:

$$V_{C_1} = R_2\, I_{L_2} + L_2 \frac{\mathrm{d}I_{L_2}}{\mathrm{d}t} + V_{C_2}. \tag{7}$$

Appendix A.3 State-space form (A, B, C, D)

Solving (4)–(7) for the time-derivatives of the state vector gives

$$\dot{V}_{C_1} = \frac{1}{C_1}\left(I_{L_1} - I_{L_2} - \tfrac{V_{C_1}}{R_x}\right), \tag{8}$$

$$\dot{V}_{C_2} = \frac{1}{C_2}\left(I_{L_2} - \tfrac{V_{C_2}}{R_{\mathrm{load}}}\right), \tag{9}$$

$$\dot{I}_{L_1} = \frac{1}{L_1}(V_s - R_1 I_{L_1} - V_{C_1})\,, \tag{10}$$

$$\dot{I}_{L_2} = \frac{1}{L_2}(V_{C_1} - R_2 I_{L_2} - V_{C_2})\,. \tag{11}$$

Reading off the coefficients yields $\dot{\mathbf{x}} = A\mathbf{x} + Bu$, $y = C\mathbf{x}$ with

$$A = \begin{bmatrix} -\frac{1}{R_x C_1} & 0 & \frac{1}{C_1} & -\frac{1}{C_1} \\ 0 & -\frac{1}{R_{\text{load}} C_2} & 0 & \frac{1}{C_2} \\ -\frac{1}{L_1} & 0 & -\frac{R_1}{L_1} & 0 \\ \frac{1}{L_2} & -\frac{1}{L_2} & 0 & -\frac{R_2}{L_2} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{L_1} \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}, \quad D = 0. \tag{12}$$

The headline entry A_{11}.

$$A_{11}(\alpha, R_x) = -\frac{1}{R_x C_1} = -\frac{1}{R_x\, C'\, \alpha\, \ell}. \tag{13}$$

For $\alpha \in (0,1)$ and $R_x \in (0,\infty)$ both factors in the denominator are smooth and strictly positive, so

$$\frac{\partial A_{11}}{\partial \alpha} = \frac{1}{R_x\, C'\, \alpha^2\, \ell}, \qquad \frac{\partial A_{11}}{\partial R_x} = \frac{1}{R_x^2\, C'\, \alpha\, \ell}, \tag{14}$$

both of which are continuous (in fact C^∞) on the open operating set $(0,1) \times (0,\infty)$. This is the differentiability property used by the WP2.2 closed-form gradient.

Appendix A.4 Closed-form transfer function $H(j\omega_0; \alpha, R_x)$

For a single-frequency analysis at $\omega = \omega_0$ the input-output transfer is

$$H(j\omega; \alpha, R_x) = C\,(j\omega\, I_4 - A)^{-1}\, B + D. \tag{15}$$

$D = 0$ for the present output choice, so H reduces to $C(j\omega I - A)^{-1}B$.

By Cramer's rule on $j\omega I - A$, H is a rational function of $j\omega$ whose numerator and denominator are polynomials of degree ≤ 4 in $j\omega$ with coefficients $\{a_k(\alpha, R_x)\}$ that are smooth on the operating envelope. The full algebraic expansion is unwieldy; the canonical evaluator is the linear solve `X = (jw*I - A) \B`, executed in both runtimes:

- MATLAB: `matlab/faultloc_pi_state_space.m` followed by `H = C * ((1j*omega*I - A) \B) + D;`
- Python: `models/faultloc_pi_section_model.py`, function `H_model(alpha, Rx, omega)`.

Cross-runtime agreement is enforced by `tests/test_pi_model_python_vs_matlab.py` against the golden CSV `tests/data/H_golden.csv`, regenerable from either runtime; max abs error $< 10^{-9}$ on a 5-cell grid (WP0.5 acceptance).

Appendix A.5 Symbolic $\partial H/\partial\alpha$ and $\partial H/\partial R_x$

Because $A(\alpha, R_x)$ is differentiable on the operating envelope, so is $H = C(j\omega I - A)^{-1}B$. Differentiating under the inverse (using $\partial_\theta M^{-1} = -M^{-1}\,(\partial_\theta M)\, M^{-1}$):

$$\frac{\partial H}{\partial \theta} = C\,(j\omega I - A)^{-1} \Big(\partial_\theta A\Big)\, (j\omega I - A)^{-1}\, B, \qquad \theta \in \{\alpha, R_x\}. \tag{16}$$

The symbolic forms of $\partial H/\partial\alpha$ and $\partial H/\partial R_x$ are produced by the MATLAB Symbolic Math Toolbox in `matlab/derive_partials.m` and emitted as self-contained MATLAB functions at `matlab/dH_dalpha.m` and `matlab/dH_dRx.m`. The script:

1. Builds A, B, C symbolically in $(\alpha, R_x, \omega, R', L', C', \ell, R_{\text{load}})$.

2. Computes $H = C(j\omega I - A)^{-1}B$ via `simplify`.
3. Takes `diff` with respect to α and R_x, applies `simplify` again.
4. Substitutes the SI defaults ($R' = 0.0728\,\Omega/\text{km}$, $L' = 0.927\times10^{-3}\,\text{H/km}$, $C' = 11.6\times10^{-9}\,\text{F/km}$, $\ell = 100\,\text{km}$, $R_{\text{load}} = 1\times10^{6}\,\Omega$).
5. Emits two callable functions of (α, R_x, ω) via `matlabFunction(..., 'Optimize', true)`.

Cross-validation against a 1×10^{-6} central finite-difference at three operating points is performed by `matlab/tests/test_partials.m`; the test passes when the relative error is below 10^{-3} at every point (in practice the auto-generated symbolic form agrees with the FD to $\sim 10^{-12}$).

Use sites. The closed-form gradient is consumed by:

- the gradient solver of § IV (`inverse_estimation/faultloc_two_stage_optimiser`, WP2.4 swap from central FD to analytic);
- the Fisher information matrix of § VIII (`inverse_estimation/faultloc_crlb_proper`, WP1.6 corrected CRLB).

Appendix A.6 Dimensional check

For the IEEE-30-style 11 kV / 100 km feeder with conductor geometry typical of medium-voltage overhead lines (Saha 2010, Springer Table 3.1, page 87), the per-km values are

$$R' = 0.0728\,\Omega/\text{km}, \qquad L' = 0.927\times10^{-3}\,\text{H/km}, \qquad C' = 11.6\times10^{-9}\,\text{F/km}. \tag{17}$$

The legacy v1 manuscript heuristics "$L' = 4R'$" and "$C' = 3R'$" are dimensionally heterodox and have been replaced by the explicit SI values above (Wang–Lopes-2023 EPSR review, §prior-art conventions, also requires SI form). Assembling section-1 quantities at $\alpha = 0.5$:

$$R_1 = 0.0728\,\Omega/\text{km}\times 50\,\text{km} = 3.64\,\Omega, \tag{18}$$
$$L_1 = 0.927\times10^{-3}\,\text{H/km}\times 50\,\text{km} = 46.35\,\text{mH}, \tag{19}$$
$$C_1 = 11.6\times10^{-9}\,\text{F/km}\times 50\,\text{km} = 0.58\,\mu\text{F}. \tag{20}$$

With $R_x = 1\,\text{k}\Omega$ this places the headline pole at

$$A_{11}(\tfrac{1}{2}, 1\,\text{k}\Omega) \;=\; -\frac{1}{R_xC_1} \;=\; -\frac{1}{1\times10^{3}\,\Omega\times0.58\times10^{-6}\,\text{F}} \;\approx\; -1724\,\text{s}^{-1}, \tag{21}$$

a fast eigenvalue compatible with the chosen sampling rate $F_s = 10\,\text{kHz}$ ($N_s = 200$, one-cycle window).

Units sanity. $[R'][\ell] = \Omega/\text{km}\cdot\text{km} = \Omega$; $[L'][\ell] = \text{H/km}\cdot\text{km} = \text{H}$; $[C'][\ell] = \text{F/km}\cdot\text{km} = \text{F}$; all A entries are in [1/s], $[B]$ is in [1/H], $[C]$ is dimensionless and the resulting $H = I/V$ is in [1/Ω]. This matches the single-bin DFT admittance H_{meas} used throughout § III.

End of Appendix A.